IILM UNIVERSITY

Greater Noida

QUESTION BANK

Programming in C

UNIT-1

20 Questions each for short and long questions

Short Questions (2 Marks)

Question
1. What is a computer? State any four basic characteristics of a computer.
2. List the major applications of computers in different fields.
3. Classify computers based on size and processing capability with examples.
4. What are the basic functional units of a computer?
5. Define memory hierarchy and list its different levels.
6. What is an Operating System? State its major functions.
7. What is a CPU? Name and explain its major functional components.
8. What is a programming language? List the different generations of programming languages.
9. Define an algorithm. State the characteristics of a good algorithm.
10. What is pseudocode? How is it different from a flowchart?
11. Differentiate between a compiler and an interpreter with examples.
12. What is the purpose of linking and loading in program execution?
13. Differentiate between source code, object code, and executable code.
14. List the major stages involved in developing a software program.
15. What is structured programming? State its advantages.
16. What are C tokens? Name their major categories.
17. Differentiate between keywords and identifiers in C.
18. What are variables and constants in C?
19. What is operator precedence and associativity in C?
20. Differentiate between implicit and explicit type conversion in C.

Long Questions (6 Marks)

Question
1. Explain the basic characteristics and major applications of computers in different fields.
2. Explain the classification of computers based on size and processing capability with suitable examples.
3. Explain the anatomy/functional units of a computer with a neat diagram.
4. Explain memory hierarchy and its different levels in detail.
5. Explain the major functions of an Operating System.
6. Explain the basic operational working of a CPU.
7. Explain the different generations and classification of programming languages.
8. Explain pseudocode and flowcharts as methods of representing an algorithm, with suitable examples.
9. Explain the difference between compiling, interpreting, linking, and loading a program.
10. Explain the complete journey of a C program from source code to execution.
11. Explain the various stages involved in developing a software program, from problem definition to testing and debugging.
12. Write an algorithm to find the largest of three numbers and explain each step.
13. Draw a flowchart to determine whether a given number is even or odd.

14. Develop an algorithm and flowchart to calculate the factorial of a given number.
15. Explain the concept of structured programming and discuss its advantages.
16. Explain the basic structure and different parts of a C program with a suitable example.
17. Explain C tokens, including keywords, identifiers, constants, and their examples.
18. Explain C data types, variables, and constants. Also discuss the rules for declaring identifiers and variables.
19. Explain operator precedence and associativity in C. Demonstrate their effect by evaluating suitable expressions step-by-step.
20. Explain implicit and explicit type conversion in C with suitable examples. Also write an algorithm and flowchart to calculate a student's grade based on marks.

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QUESTION BANK

Programming in C

Unit-2

20 Questions each for short and long questions

Section A: Short Answer Questions

Question
1. What is input and output in C?
2. What is the purpose of printf() and scanf()?
3. What are format specifiers in C? Give examples.
4. What are escape sequences? Give examples.
5. What is a control statement? Mention its categories.
6. Explain the if statement with its syntax.
7. What is an if-else statement?
8. What is a nested if statement?
9. What is an if-else-if ladder?
10. What is the conditional operator (?:)?
11. What is a switch statement?
12. What is the difference between while, for, and do-while loops?
13. What is the purpose of the break statement?
14. What is the purpose of the continue statement?
15. What are goto, return, and exit() statements?

Section B: Long Answer / Programming Questions

Question
1. Explain printf() and scanf() in detail with suitable C programs and commonly used format specifiers.
2. Explain the if, if-else, nested if, and if-else-if ladder statements with syntax and suitable examples.
3. Write a C program to check whether a given number is positive, negative, or zero using if-else.
4. Write a C program to check whether a given number is even or odd.
5. Write a C program to find the largest among three numbers using nested if.
6. Write a C program to check whether a given year is a leap year.
7. Write a C program to calculate the grade of a student based on marks using an if-else-if ladder.
8. Explain the conditional operator (?:) with syntax and a suitable example. Compare it with the if-else statement.
9. Explain the switch statement with syntax and a suitable example. Write a C program to display the day of the week using switch.
10. Write a C program to implement a simple calculator using switch-case.
11. Explain and compare the for, while, and do-while loops with suitable examples.
12. Write a C program to print the multiplication table of a given number using a for loop.
13. Write a C program to find the sum of the first n natural numbers using a while loop and explain its working.
14. Write a C program to determine whether a given number is prime or not using a loop.
15. Write a menu-driven C program using switch to perform the following operations: 1 – Check Even/Odd, 2 – Check Positive/Negative, 3 – Find Square, 4 – Find Cube.

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QUESTION BANK

Programming in C

Unit- 3(1-D Array)

20 Questions each for short and long questions

Section A: Short Answer Questions

Question
1. Write a program to calculate the sum and average of elements in a 1D array.
2. Write a syntax to declaring a one-dimensional array. Give an example.
3. Write a program to find the largest element in a 1D array.
4. Write a program to count the even and odd elements in a 1D array.
5. How can i access the first and last element in the array? Give me an example.

Section B: Long Answer / Programming Questions

Question
1. WAP to read N integer into a 1-D array & find the sum and largest element of the array.
2. WAP to read N integer into N integers into an array and count how many elements are positive and how many are negative
3. WAP to read N elements into a 1-D array and find the sum and average of the element.